

By defn. of existential quantifier, $\exists w((w \neq a_1) \wedge (w \neq a_2) \wedge (w \neq a_3))$ is False.
 By defn. of universal quantifier, $\forall z \exists w((w \neq a_1) \wedge (w \neq a_2) \wedge (w \neq z))$ is False.
 By defn. of universal quantifier, $\forall y \forall z \exists w((w \neq a_1) \wedge (w \neq y) \wedge (w \neq z))$ is False.
 By defn. of universal quantifier, $\forall x \forall y \forall z \exists w((w \neq x) \wedge (w \neq y) \wedge (w \neq z))$ is False.

Direction 2: $\forall x \forall y \forall z \exists w((w \neq x) \wedge (w \neq y) \wedge (w \neq z))$ is False $\Rightarrow$ Common domain consists of at most 3 elements.

By defn. of negation, $\neg \forall x \forall y \forall z \exists w((w \neq x) \wedge (w \neq y) \wedge (w \neq z))$ is True.
 $\neg \forall x \forall y \forall z \exists w((w \neq x) \wedge (w \neq y) \wedge (w \neq z)) \equiv \exists x \neg \forall y \forall z \exists w((w \neq x) \wedge (w \neq y) \wedge (w \neq z))$ (De Morgan's law of quantifiers) $\equiv \exists x \exists y \neg \forall z \exists w((w \neq x) \wedge (w \neq y) \wedge (w \neq z))$ (De Morgan's law of quantifiers) $\equiv \exists x \exists y \exists z \neg \exists w((w \neq x) \wedge (w \neq y) \wedge (w \neq z))$ (De Morgan's law of quantifiers) $\equiv \exists x \exists y \exists z \forall w \neg((w \neq x) \wedge (w \neq y) \wedge (w \neq z))$ (De Morgan's law of quantifiers) $\equiv \exists x \exists y \exists z \forall w(\neg(w \neq x) \vee \neg(w \neq y) \vee \neg(w \neq z))$ (1st De Morgan's law) $\equiv \exists x \exists y \exists z \forall w((w = x) \vee (w = y) \vee (w = z))$ is True. By defn. of existential quantifier, there exists an element a_1 in the domain s.t. $\exists y \exists z \forall w((w = a_1) \vee (w = y) \vee (w = z))$ is True. By defn. of existential quantifier, there exists an element a_2 in the domain s.t. $\exists z \forall w((w = a_1) \vee (w = a_2) \vee (w = z))$ is True. By defn. of existential quantifier, there exists an element a_3 in the domain s.t. $\forall w((w = a_1) \vee (w = a_2) \vee (w = a_3))$ is True. By defn. of universal quantifier, for each element b in the domain, $((b = a_1) \vee (b = a_2) \vee (b = a_3))$ is True. By defn. of OR, at least one of $(b = a_1), (b = a_2), (b = a_3)$ must be True. If a_1, a_2, a_3 are all distinct elements, and b is different from them all, $(b = a_1), (b = a_2), (b = a_3)$ are all False respectively. $\therefore$ There must be less than 4 elements in the domain for having at least one of $(b = a_1), (b = a_2), (b = a_3)$ to be True.

$\therefore$ Common domain consists of at most 3 elements $\square$